

Experiment 6: Prompt Strategy Design – Implement and Compare the Effectiveness of Zero-shot, One-shot, and Few-shot Prompts

Aim

To implement and compare the effectiveness of Zero-shot, One-shot, and Few-shot prompting techniques using the OpenAI API.

Python Code:

```
from openai import OpenAI

# Paste your OpenAI API Key here
client = OpenAI(
    api_key="sk-proj-FboLgW768fa2IYU2-KXDkmHy_cK9JgnW_P9A-A_3SSStE-
Fm94OYnHlDpB6mo5qbWfba9c7LnGdT3BlbkFJoqosff_LYE2VyPCdbNGX_4p3ghTZhcd
Bt4Ff_Fh0tYqmNnhjB64OmSAx8KtGx7X9XtRDtN0cUA"
)

task = "Explain the importance of Artificial Intelligence in education in about 100 words."

# ----- Zero-shot Prompt -----
zero_shot = task

response = client.responses.create(
    model="gpt-5.4-mini",
    input=zero_shot
)

print("===== ZERO-SHOT OUTPUT =====")
print(response.output_text)

# ----- One-shot Prompt -----
one_shot = ""
Example:

Question:
Explain the importance of Artificial Intelligence in healthcare in about 100 words.

Answer:
Artificial Intelligence improves healthcare by assisting doctors in disease diagnosis,
medical imaging, personalized treatment, robotic surgery, and patient monitoring.
It increases accuracy, reduces workload, and improves patient care through faster
and more reliable medical decisions.
```

Now answer the following:

Explain the importance of Artificial Intelligence in education in about 100 words.

"""

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=one_shot  
)
```

```
print("\n===== ONE-SHOT OUTPUT =====")  
print(response.output_text)
```

----- Few-shot Prompt -----

few_shot = """

Example 1

Question:

Explain the importance of Artificial Intelligence in banking.

Answer:

AI improves banking by detecting fraud, automating customer service, analyzing financial risks, and providing personalized banking experiences.

Example 2

Question:

Explain the importance of Artificial Intelligence in agriculture.

Answer:

AI helps farmers monitor crops, predict weather, detect plant diseases, and improve crop production using smart technologies.

Now answer the following:

Explain the importance of Artificial Intelligence in education in about 100 words.

"""

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=few_shot  
)
```

```
print("\n===== FEW-SHOT OUTPUT =====")
```

```
print(response.output_text)
```

Output:

===== ZERO-SHOT OUTPUT =====

Artificial Intelligence is becoming very important in education because it helps make learning more personal, efficient, and accessible. AI can adjust lessons to match each student's level, helping them learn at their own pace. It also gives teachers useful data about student progress, making it easier to identify strengths and weaknesses. AI tools like chatbots, automated grading, and smart tutoring systems save time and support both teachers and learners. In addition, AI can help students with disabilities through speech recognition, translation, and other assistive technologies. Overall, AI improves the quality of education and prepares students for a technology-driven future.

===== ONE-SHOT OUTPUT =====

Artificial Intelligence is important in education because it makes learning more personalized, efficient, and accessible. AI can analyze a student's strengths and weaknesses and suggest lessons, exercises, or practice at the right level. It helps teachers save time by automating tasks like grading, attendance, and feedback. AI-powered tools such as chatbots and virtual tutors can support students anytime, making learning available beyond the classroom. It also helps identify students who need extra help early. Overall, Artificial Intelligence improves teaching, supports learners, and creates a smarter education system.

===== FEW-SHOT OUTPUT =====

Artificial Intelligence is important in education because it helps make learning more personalized, efficient, and accessible. AI can analyze each student's strengths and weaknesses and recommend lessons, exercises, or feedback suited to their needs. It also supports teachers by automating tasks such as grading, attendance, and answering common questions, saving time for more meaningful teaching. AI-powered tools like virtual tutors and smart learning platforms can assist students anytime, anywhere. In addition, AI can help identify students who are struggling early, allowing timely support. Overall, AI improves the quality and effectiveness of education.

Result

The Zero-shot, One-shot, and Few-shot prompting techniques were successfully implemented using the OpenAI API. The responses generated by each strategy were compared based on accuracy, relevance, response quality, and overall effectiveness. The Few-shot prompting strategy produced the most accurate, relevant, and well-structured response, demonstrating the highest overall effectiveness.